Bitcoin Target and Leading Zeros Explained

In Bitcoin mining, the target is derived from a compact representation (nBits) using the formula:

hashTarget = mantissa × 2^(8 × (exponent – 3))

A valid block hash must be less than or equal to this target. The smaller the target, the more leading zeros are required in the hash.

Example 1: nBits = 0x1d00ffff

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Exponent = 0x1d = 29
Mantissa = 0x00ffff

Target = 65535 × 2^(8 × (29 – 3))
       = 65535 × 2^208

Hex form:
00000000FFFF0000000000000000000000000000000000000000000000000000

Leading zeros:
8 hex zeros = 32 bits
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Example 2: nBits = 0x1705c739

```
Exponent = 0x17 = 23
Mantissa = 0x05c739

Target = 0x05c739 × 2^(8 × (23 – 3))
       = 0x05c739 × 2^160

Hex form:
0000000000000000000005c7390000000000000000000000000000000000000000

Leading zeros:
18 hex zeros = 72 bits
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Summary

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1. Extract exponent and mantissa from nBits.
2. Compute target using the formula.
3. Convert target to 256-bit hex.
4. Count leading zeros (each hex digit = 4 bits).

Smaller target → more leading zeros → higher mining difficulty.
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